

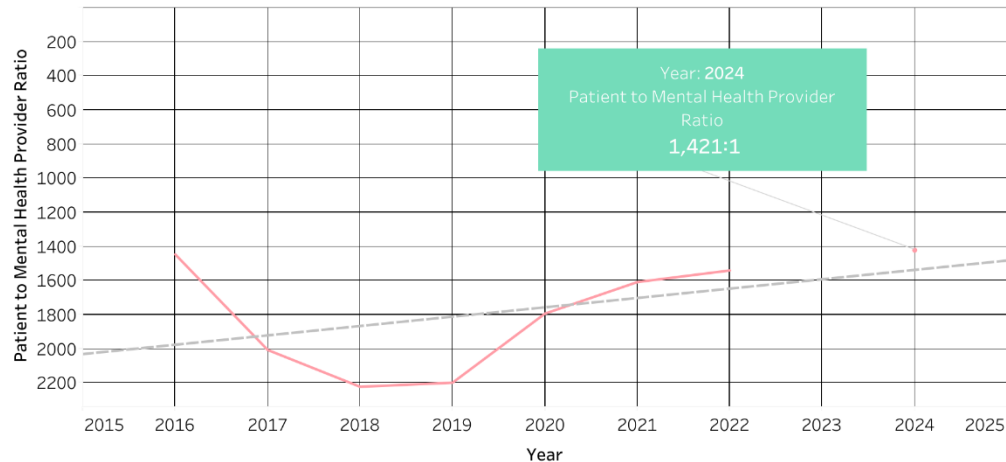
A Peace And Safety Centered Analysis Of Education, Mental Health Provider Accessibility, And Food Insecurity In Edgecomb County

This document will record the methods, analysis, and visualizations produced in correspondence with Biokind to help communicate various notable metrics, useful insights, and possible solutions to the Rural Opportunity Institute.

Initial Discoveries:

Figure 1.

Patient to Mental Health Provider Ratio Over Time (Patient #:1) (2016 - 2024)

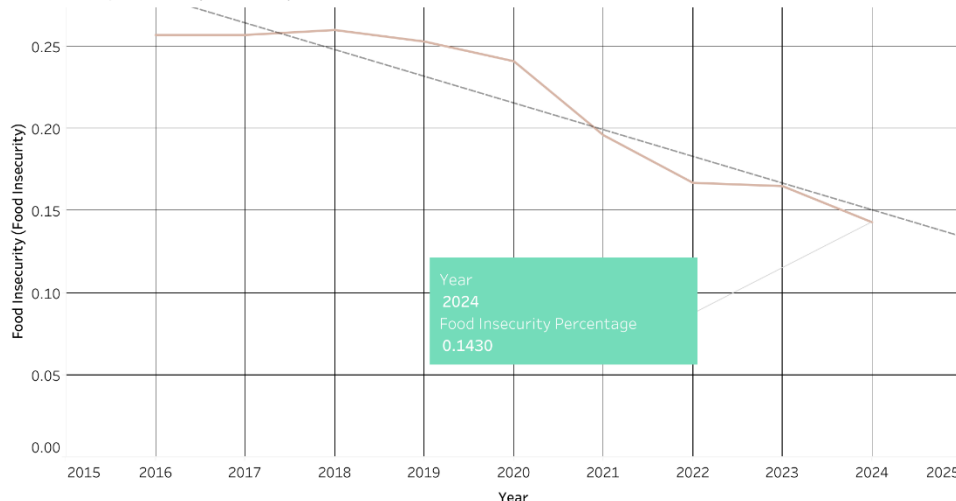


Something notable was that mental health providers were on the decline from 2016 to 2018 but if you observe the mental health ratios in Figure 1 they have been improving since 2019. Mental health is important to consider since regardless of what factors affect the everyday life of people in Edgecomb

county, the mental health of any given person is affected by any social determinants like food insecurity, lack of transportation, and intimate partner violence. However, knowing mental health accessibility fails to address which of those social determinants/factors are most prevalent and in what forms they might take so exploration of relevant factors seems appropriate.

Figure 2.

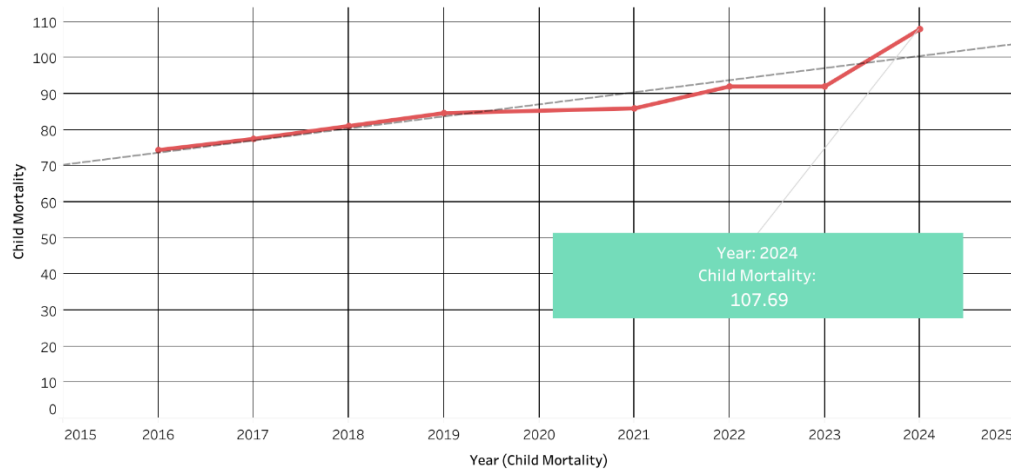
Food Insecurity Over Time (2016 - 2023)



In Figure 2 the percent of the population experiencing food insecurity has been on a steady decline since 2018. This means that lack of food access is unlikely to be the reason for an increase in demand/need for mental health support.

Figure 3.

Number of deaths among residents under age 18 per 100,000 population 2016 - 2024



If you observe Figure 3 there has been a rise in child mortality that has been occurring for nearly a decade in Edgecomb County. When a family goes through a death it can be devastating. Whole communities can experience shock discovering a neighbor has passed

on. This can certainly increase demand for mental health resources and be one of many different reasons we have observed shifts in mental health accessibility. As for the child deaths themselves, one of the possible reasons it is on an incline is because of poor dependent finances and employment. Hence, exploring relevant data seemed pertinent.

Figure 4.

Growforce Groth by Collar Type For The Years 2016 -2023

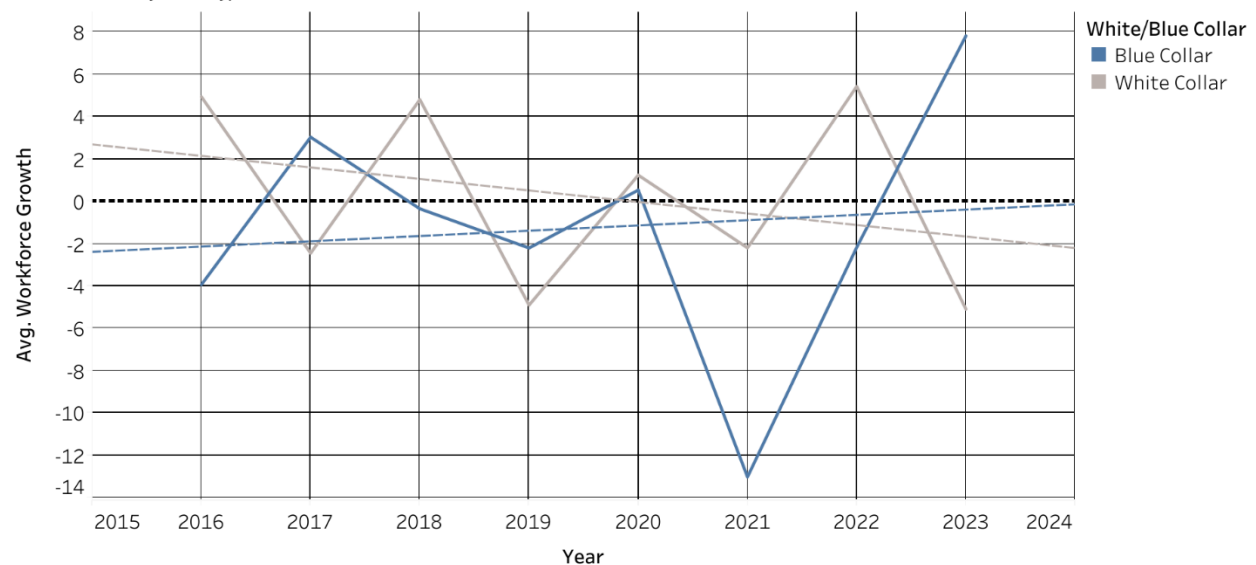
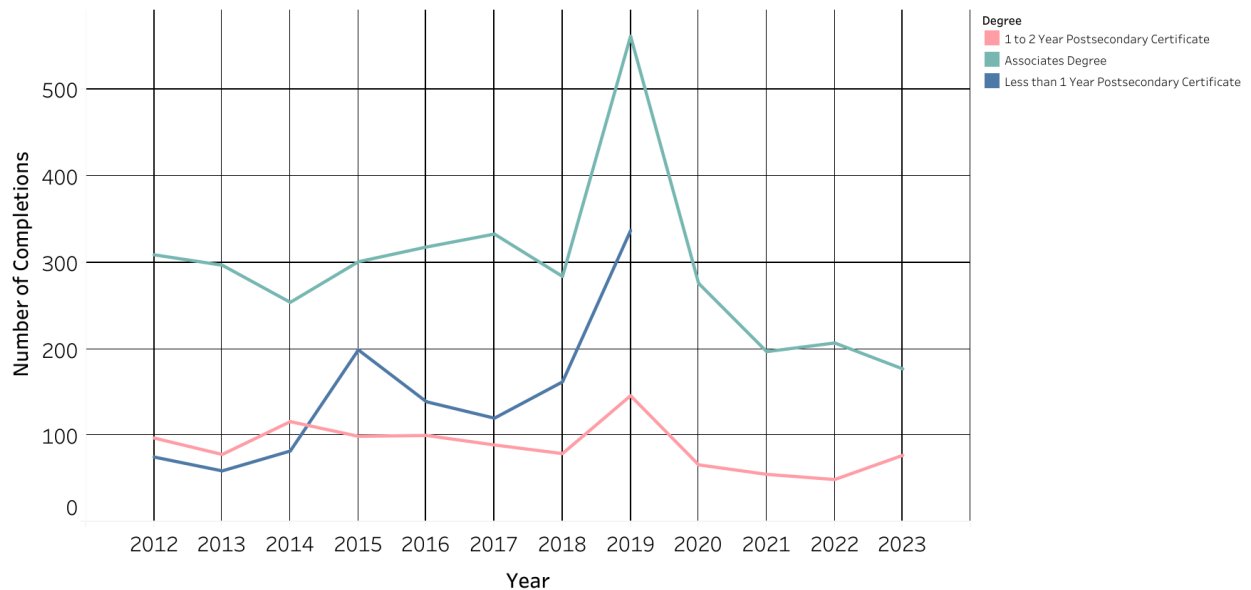


Figure 4 shows that white collar job employment growth has generally been on decline while blue collar jobs, while being extremely variable, have generally been on the incline. Many years, the employment growth for blue collar and/or white collar is in the negative. This tells us that the amount of people losing their jobs outweighs those who are able to find jobs. A extremely notable year for blue collar jobs was 2021 which was flagged by a z-score outlier test ($z\text{-score} = -2.22$ or roughly 2 standard deviations below the mean). That very same year, white collar employment growth was negative. In summary, employment growth is inconsistent and poor. It could be that very inconsistency that contributes to child mortality because those unemployed might not be able to provide to their children. Higher education has ties to employability, especially for white collar jobs. Certification accessibility could definitely contribute to more blue collar job hires and therefore greater employment growth. Education seems important to understanding possible insights and solutions in relation to employment growth.

Methods and Models:

Figure 5.

Number of Certificates Awarded by Award Type Over Time 2012 -2023



Before taking a closer look at the relationship between education and workforce growth it is important to address and uncover multicollinearity. Looking at the data DATAUSA has on Edgecomb County for its awarded degree in Figure 5, the shapes of the line plots are very similar. The Variance Inflation Factor (VIF) reflect this. Typically, a VIF of 5 or greater for a variable indicates multicollinearity. What the observed VIFs indicate in this context is that when any particular amount of one degree is awarded the amount awarded in the other degree types reflect that increase. In other words, award amounts of each award type are correlated.

	Variable	VIF
0	Associates Degree	46.201260
1	1 to 2 Year Postsecondary Certificate	26.164925
2	Less than 1 Year Postsecondary Certificate	14.701305

Therefore, before we create an Ordinary Least Squares (OLS) we must break down each variable into principle components (PCs). Ideally we can create enough PCs so that, at the very least, 95% of the variance is explained. Here were the automatic weights created:

Principle Component 1.

```
Associates Degree      0.640033
1 to 2 Year Postsecondary Certificate  0.565825
Less than 1 Year Postsecondary Certificate  0.519808
Name: PC1>Loading, dtype: float64
```

Principle Component 2.

```
Less than 1 Year Postsecondary Certificate  0.777126
1 to 2 Year Postsecondary Certificate -0.624342
Associates Degree -0.079195
Name: PC2>Loading, dtype: float64
```

PCA-OLS.

Explained Variance Ratio: [0.7582331 0.20505426]					
Total Variance Explained: 0.9632873579394454					
OLS Regression Results					
=====					
Dep. Variable:	Workforce Growth	R-squared:	0.704		
Model:	OLS	Adj. R-squared:	0.620		
Method:	Least Squares	F-statistic:	8.336		
Date:	Sun, 05 Apr 2026	Prob (F-statistic):	0.0141		
Time:	01:53:30	Log-Likelihood:	-22.124		
No. Observations:	10	AIC:	50.25		
Df Residuals:	7	BIC:	51.16		
Df Model:	2				
Covariance Type:	nonrobust				
=====					
	coef	std err	t	P> t	[0.025 0.975]

const	0.7784	0.836	0.931	0.383	-1.198 2.754
PC1	-0.5714	0.554	-1.031	0.337	-1.882 0.739
PC2	-4.2097	1.066	-3.951	0.006	-6.729 -1.690
=====					
Omnibus:		0.949	Durbin-Watson:	2.577	
Prob(Omnibus):		0.622	Jarque-Bera (JB):	0.644	
Skew:		-0.545	Prob(JB):	0.725	
Kurtosis:		2.404	Cond. No.	1.92	
=====					

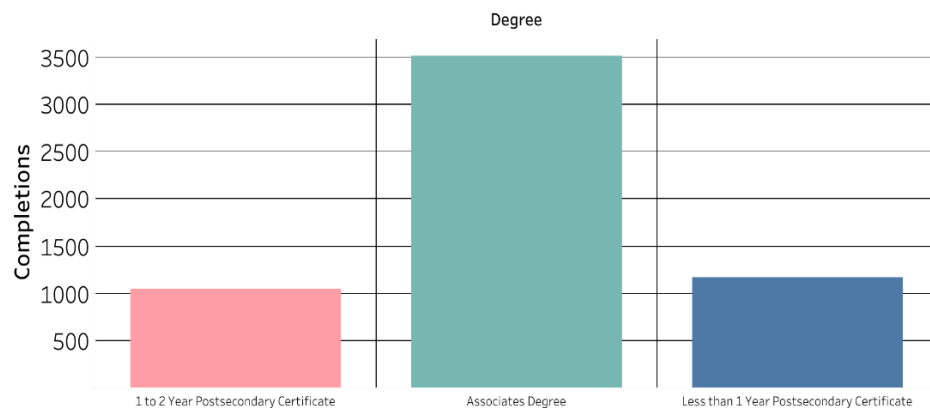
Overall, the model performed well, the R-squared of 0.7 means the model could correctly predict on the data fed to it 70% of the time. From the OLS we discover that Principle component 2 is statistically significant as a predictor of the percentage of workforce growth. PC2 has a negative coefficient therefore it had a negative effect on workforce growth. However, "1 to 2 Postsecondary Certificate" weighed in negatively into that component thereby implying it had a significant positive affect on workforce growth. While "Less than 1 Year Postsecondary Certificate" weighed in very positively implying that it actually had a very negative affect on workforce growth.

Insights and Suggestions:

To give general overview, we wanted to explore ways to mitigate mental care need by improving child mortality by finding some solution to the variability of workforce growth. While the observed points between these topics are loosely tied and have no direct connection we were able to establish a connection between workforce growth and awarded degrees.

Figure 6.

Number of Completions Per Award Type 2012 - 2023



My personal biggest suggestion is to put more focus/initiatives into making certain degrees/certificates more accessible. From our PCA-OLS we were able to uncover that long-term degrees like "1 to 2 Postsecondary Certificate" and even "Associates Degree" had a positive relationship with workforce growth. From Figure 6 you

can see there is very large inbalance in the degrees awarded so one possible initiative is advertising and possibly collaborating with local businesses to make more long-term awards like 1 to 2 year postsecondary certificates more visible/available while being also very hands on and practical. We can refer back to figure 4 and see that when there is negative workforce growth, it more often blue collar. If long-term awards can play into that and offer abundant experience then people with those certificates will both have agency in their respective fields as well as the practical experience to match, thereby improving employability, especially in blue collar jobs which often require technical hands-on skills.